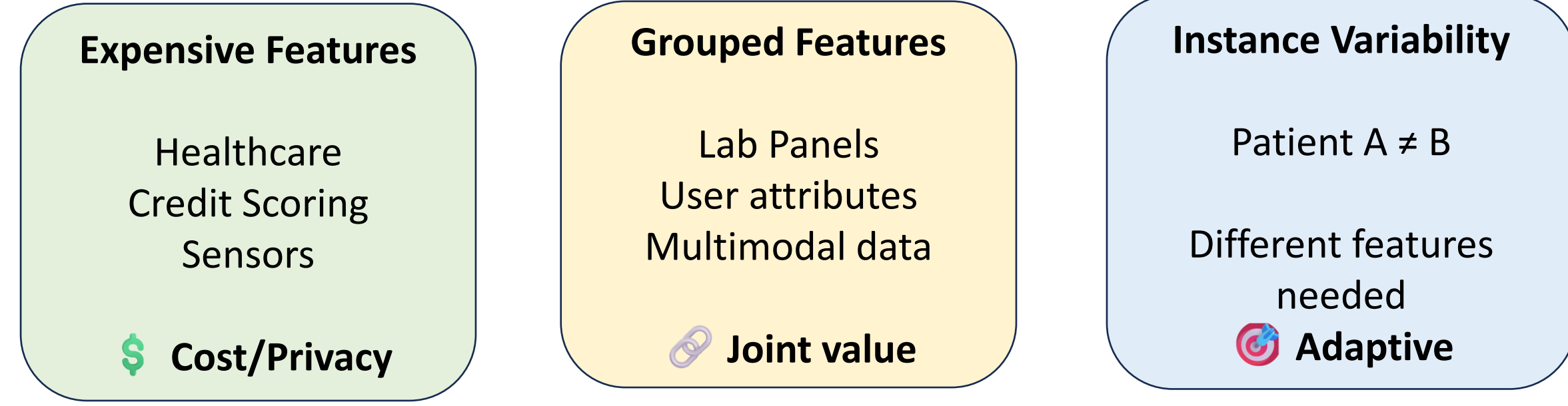


Introduction

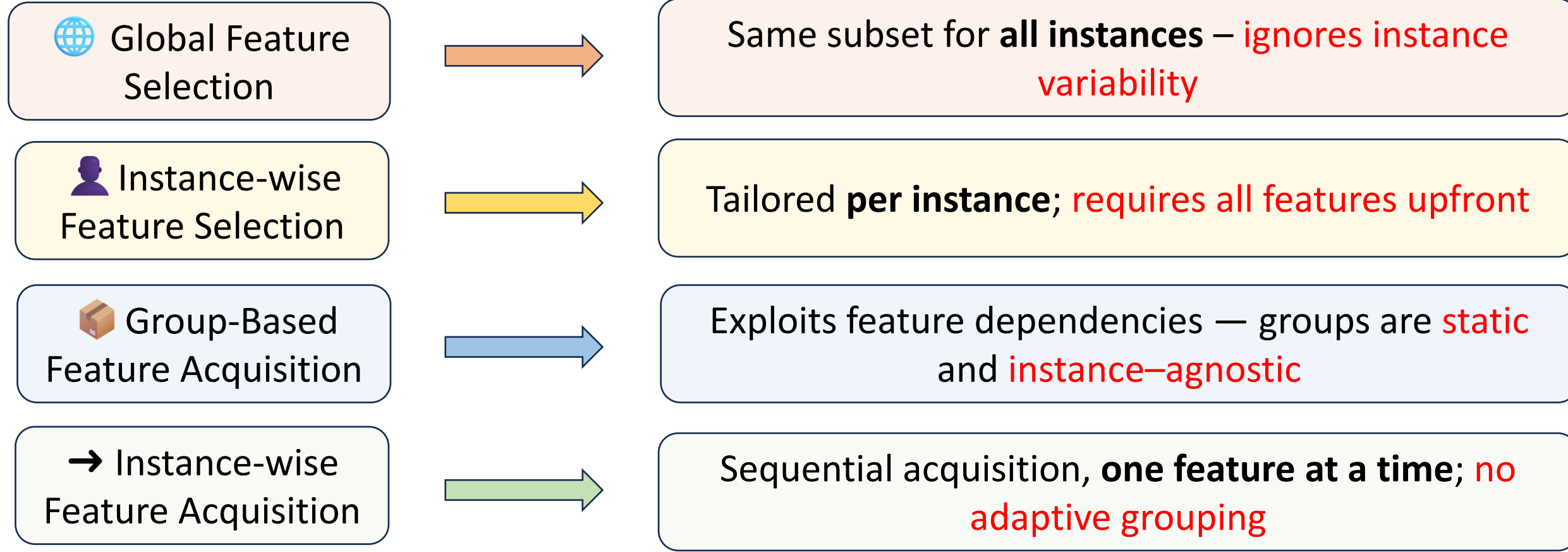


Static Feature Selection is insufficient

Sequential + Instant-wise Acquisition

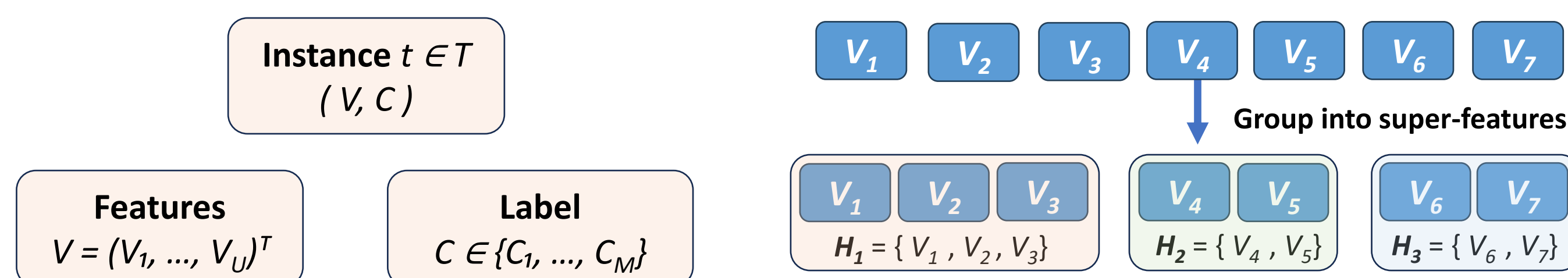
Objective: jointly determine **how many features to acquire at each step** (group size), **when to stop**, and **class label** of each instance — minimizing total expected acquisition + misclassification cost

Related Work



SeqFDM: instance-wise acquisition of **dynamically-sized feature groups** with **DP-optimal stopping**

Problem Description



Feature Cost $d_u > 0$

Misclassification Penalty $N_{kl} \geq 0$

Assumptions:

- fixed global acquisition order
- class-conditional feature independence
- max group size δ

Expected total cost (acquisition + misclassification):

$$J(\{\delta_q\}_{q=1}^Q, Q, Y_Q) = \mathbb{E} \left\{ \sum_{q=1}^Q \sum_{u=h+1}^{h+\delta_q} d_u \right\} + \sum_{l=1}^M \sum_{k=1}^M N_{kl} P(Y_Q = l, C = C_k)$$

Feature acquisition cost *Expected Misclassification cost*

Choose $\{\delta_q\}_{q=1}^Q, Q, Y_Q$ (group sizes, stopping stage, label) that **minimize J**

Solution

Posterior Update

- Belief over classes after q super-features, updated by **Bayes' rule**:

$$\zeta_q = \frac{\text{diag}(\Phi_q(H_q)) \zeta_{q-1}}{\Phi_q^T(H_q) \zeta_{q-1}}, \quad \zeta_0 = [p_1, \dots, p_M]^T$$

Optimum Classification

$$Y_Q^{\text{opt}} = \arg \min_{1 \leq l \leq M} [N_l^T \zeta_Q], \quad N_l = [N_{1l}, N_{2l}, \dots, N_{Ml}]^T$$

Optimum Stopping via DP

$$\tilde{J}_q(\zeta_q) = \min [g(\zeta_q), \tilde{A}_q^1(\zeta_q), \tilde{A}_q^2(\zeta_q), \dots, \tilde{A}_q^\delta(\zeta_q)]$$

cost of stopping $g(\zeta_q)$ vs. cost of continuing $\tilde{A}_q^z(\zeta_q)$

- Expected continuation cost for acquiring $z \in \{1, \dots, \delta\}$ more features:

$$\tilde{A}_q^z(\zeta_q) = \sum_{u=h+\delta_q+1}^{h+\delta_q+z} d_u + \sum_{H_{q+1}} \Phi_{q+1}^T(H_{q+1}) \zeta_q \tilde{J}_{q+1}(\zeta_{q+1})$$

- Next group size minimizes continuation cost:

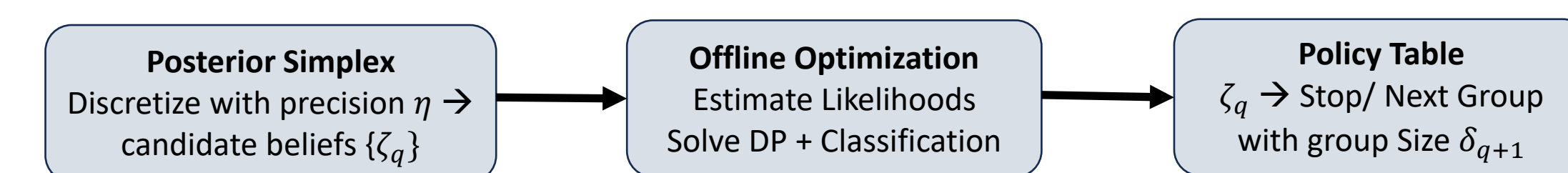
$$\delta_{q+1} = \arg \min_{z \in \{1, \dots, \delta\}} \tilde{A}_q^z(\zeta_q)$$

- Stopping rule:

$$Q^{\text{opt}} = \text{first } q < U \text{ with } g(\zeta_q) \leq \tilde{A}_q(\zeta_q); \quad \text{else } Q^{\text{opt}} = U$$

Proposed Approach: SeqFDM

Training (offline)



Testing (online)

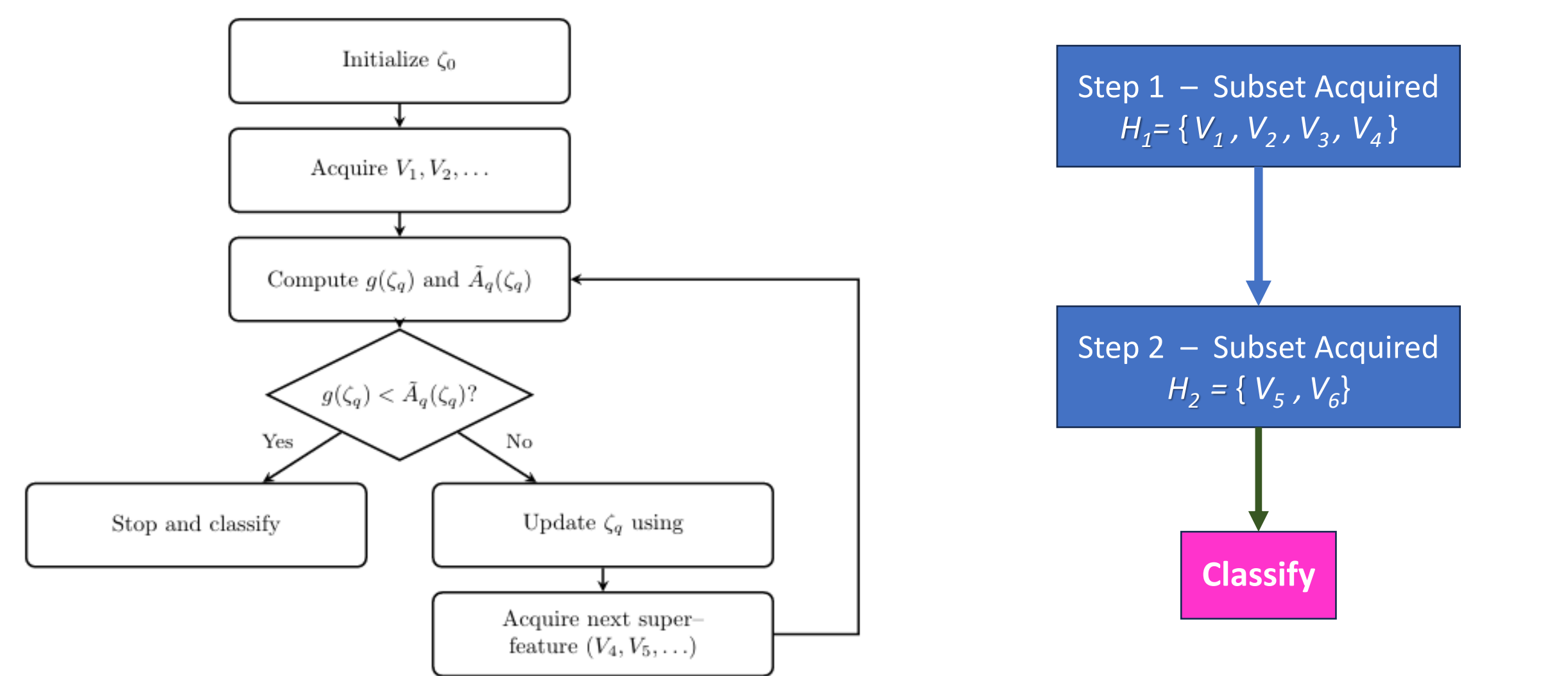


Fig: SeqFDM test-stage operation: stop-vs-acquire decisions at each step

Results

9 public datasets (MLL, Dexter, Leukemia, Prostate, Lung2, Madelon, Spambase, Eye, Diabetes)
· **5-fold CV** · metrics: accuracy, avg. #features, inference time · baselines: **ETANA**, **gl**, **GLasso** ·
Max-Variance ordering, $\delta = 3, 4$ bins, 0.1 group discount

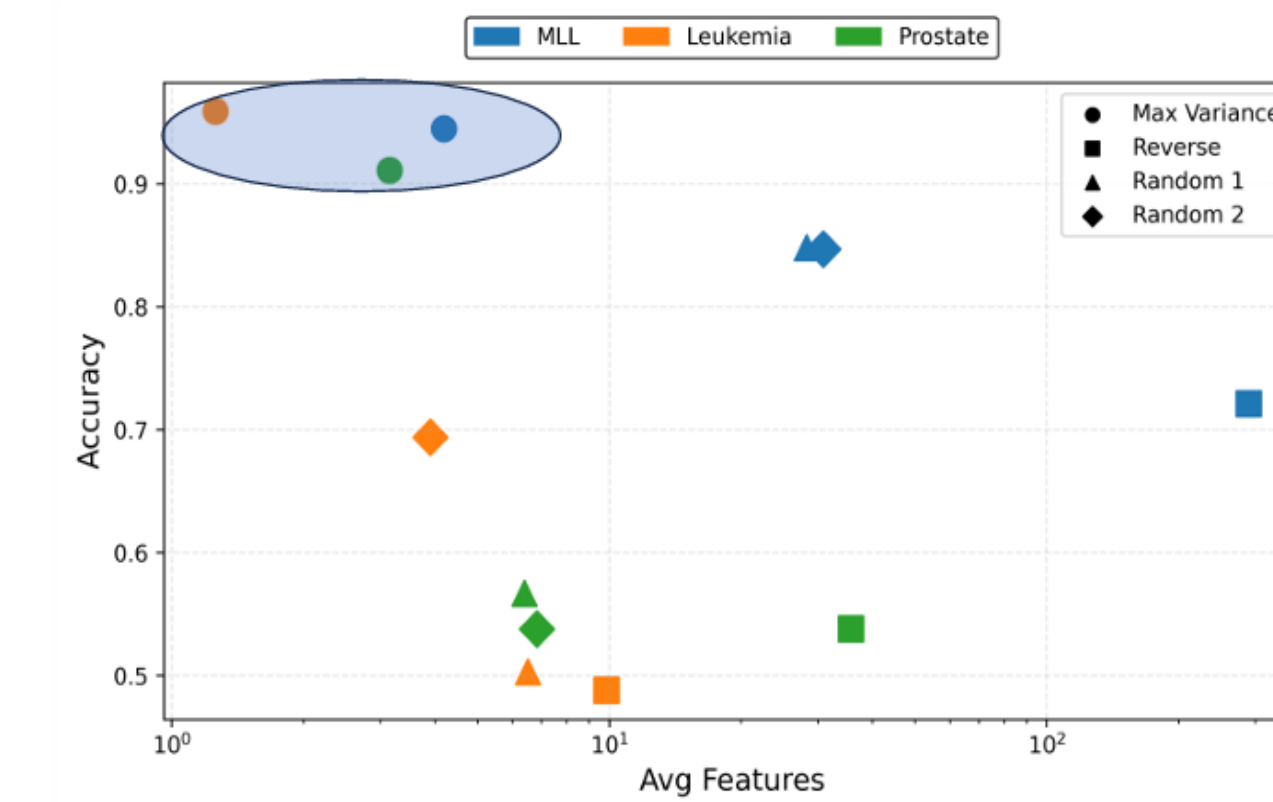


Fig: Feature ordering strategies

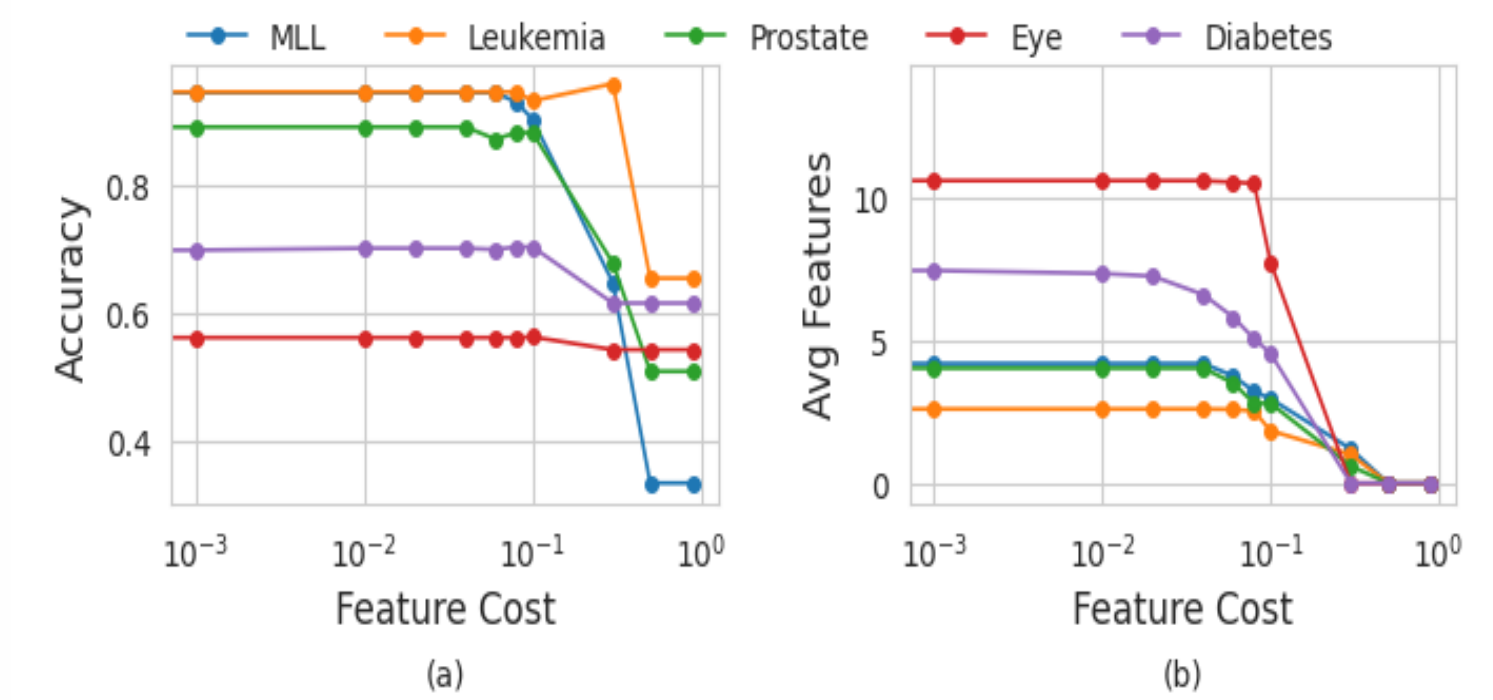


Fig: Effect of feature cost d_u

Dataset	Accuracy		# Features		Inference Time	
	SeqFDM	ETANA	SeqFDM	ETANA	SeqFDM	ETANA
MLL	0.945	0.945	4.190	4.214	0.001	0.002
DEXTER	0.843	0.767	16.843	17.854	0.010	0.022
LEUKEMIA	0.959	0.916	1.257	2.535	0.001	0.001
PROSTATE	0.911	0.873	3.150	4.068	0.002	0.002
LUNG2	0.842	0.867	17.231	16.453	0.010	0.018
MADOLON	0.618	0.616	2.092	3.642	0.018	0.053
SPAMBASE	0.746	0.651	2.176	3.014	0.042	0.077
EYE	0.562	0.560	6.114	0.804	0.230	0.119
DIABETES	0.691	0.694	3.582	3.698	0.005	0.008

^aBold gray-shading indicates the best accuracy, inference time, and the fewest features.

Tab: SeqFDM vs. ETANA: accuracy, #features, inference time

	SeqFDM	ETANA	gl	GLasso
Accuracy	0.983	0.983	0.575	1.000
# Features	3.486	3.237	85.740	8.100
Average Cost	0.039	0.037	1.082	0.163

Accuracy Gain	Feature Saving	Speedup
+4 % Leukemia Prostate	↓ +50 % Leukemia	~50 % Faster Inference
+9 % Spambase	↓ 23% Prostate	

Heterogeneous Feature Cost

98.3% Accuracy
27.7x Cost reduction Compared to gl and GLasso

vs. GLasso

↑ Accuracy **11%**
↓ Features **<0.5%** used

vs. gl

↑ Accuracy **33%**
↓ Features **99.9%** fewer

Tab: Heterogeneous feature costs (Disease Prediction)

DATA INSTANCE A										Predicted label	
SeqFDM	Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose		No	✓
	Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose		Yes	✗
gl	Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose		No	✗
	Skin Thickness	Pregnancies	Diabetes Pedigree Function	Age	BMI	Blood Pressure	Insulin	Glucose		Yes	✓

Fig: Instance-wise feature group acquisition, SeqFDM vs. gl on Diabetes, two examples

Conclusions

Takeaways

- Cost-aware, adaptive group acquisition
- Comparable accuracy, fewer acquired features
- Faster inference than prior methods

Future Work

